Assessing Nurses' Knowledge, Perception, and Practice about Pandemic Disasters: A Cross-sectional Study

Samirah Alkhaldi
ID: 446200992
Master Thesis Defense
College of Nursing
King Saud University

- **Thesis Supervisor:**
  - **Prof. Ahmad Aboshaiqah**

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Introduction

Pandemic disasters are emerging as defining challenges to global public health in the 21st century, revealing gaps across health systems in preparation for and response to pandemics.

Nurses, the largest workforce category, shoulder the greatest burden: infection prevention, patient triage, routine and urgent care, and communication with patients and families under intense psychological and physical stress.

COVID-19 showed that frontline preparedness is uneven, and that assessing the triad of **knowledge, perception and practice** is important to strengthen nurses' capacity to manage pandemic disasters.

Introduction

- Most global **MERS** cases were reported in Saudi Arabia: **2,196 of 2,605** cases by August 2023 (WHO, 2023a).
- Saudi Arabia had **243,336 nurses** across all health sectors in 2024 (MOH, 2025).
- The Taif branch of the MOH has **16 hospitals** with **2,640 beds** and **4,638 nurses and midwives** (MOH, 2025).
- National emergency planning has been strengthened, but it succeeds only through frontline nurses' **knowledge, confidence and consistent practice**.
- Previous Taif evidence (Brinjee et al., 2021) addressed **emergency nurses' training needs only**.

Key Variables:

- **Self-Perceived Knowledge**

Nurses' self-assessed familiarity, understanding and confidence regarding disaster principles, hospital plans, triage, emergency communication and nursing roles.

- **Perceptions**

Nurses' beliefs, attitudes, confidence, professional responsibility, perceived organizational support and readiness to respond.

- **Self-Reported Practice**

Reported engagement in drills, emergency equipment, evacuation, documentation, infection control, PPE, protocol review and lessons learned.

- **Demographic Characteristics**

Age, gender, marital status, educational level, years of experience and practice area.

Problem Statement

Despite nurses' contribution to pandemic preparedness and response, previous Saudi studies revealed **persistent gaps in knowledge, training and practical preparedness** among nurses concerning disasters.

However, **few data** exist on nurses' knowledge, perception and practice about pandemic disasters in **governmental hospitals in the Taif Region**. Without context-specific evidence, healthcare leaders cannot identify preparedness gaps or develop targeted education, simulations and institutional interventions.

Significance of the Study

- Little evidence exists on whether nurses in **Taif governmental hospitals** have adequate knowledge, positive perceptions and appropriate practices related to pandemic disasters.
- Previous research focused on **overall disaster preparedness, emergency nurses, or healthcare professionals as a whole**.
- If one dimension is weak, others may be limited; **knowledge may not be put into practice** without confidence, training, resources and organizational support.
- Results can highlight **education and simulation-training needs** and support hospital leaders and policy makers in designing preparedness programs.

Purpose Statement

Aim and Objectives

The aim of this study was to assess nurses' self-perceived knowledge, perceptions and self-reported practices regarding pandemic disasters among nurses working in **governmental hospitals in the Taif Region**, Saudi Arabia. Specifically, it sought to:

- Determine the level of each of the three domains.
- Examine the relationships among the three domains.
- Identify differences according to gender and educational level.

Research Questions

1. What is the level of nurses' **self-perceived knowledge** regarding pandemic disasters in the Taif Region?
2. What is the level of nurses' **perceptions** regarding pandemic disasters in the Taif Region?
3. What is the level of nurses' **self-reported practices** regarding pandemic disasters in the Taif Region?
4. What **relationships** exist among nurses' self-perceived knowledge, perceptions and self-reported practices?
5. To what extent is self-perceived knowledge **associated with** perceptions and self-reported practices?
6. Are there differences in the three domains according to **gender**?
7. Are there differences in the three domains according to **educational level**?

Conceptual Framework

Framework:

- Based on the Knowledge–Attitude/Perception–Practice (KAP) model.

Self-Perceived Knowledge:

- Nurses rate their own familiarity and confidence (not a scored test).

Perception & Practice:

- Attitudes, support and readiness; reported preparedness behaviors.

Use:

- Guided levels, relationships and differences; not a causal pathway.

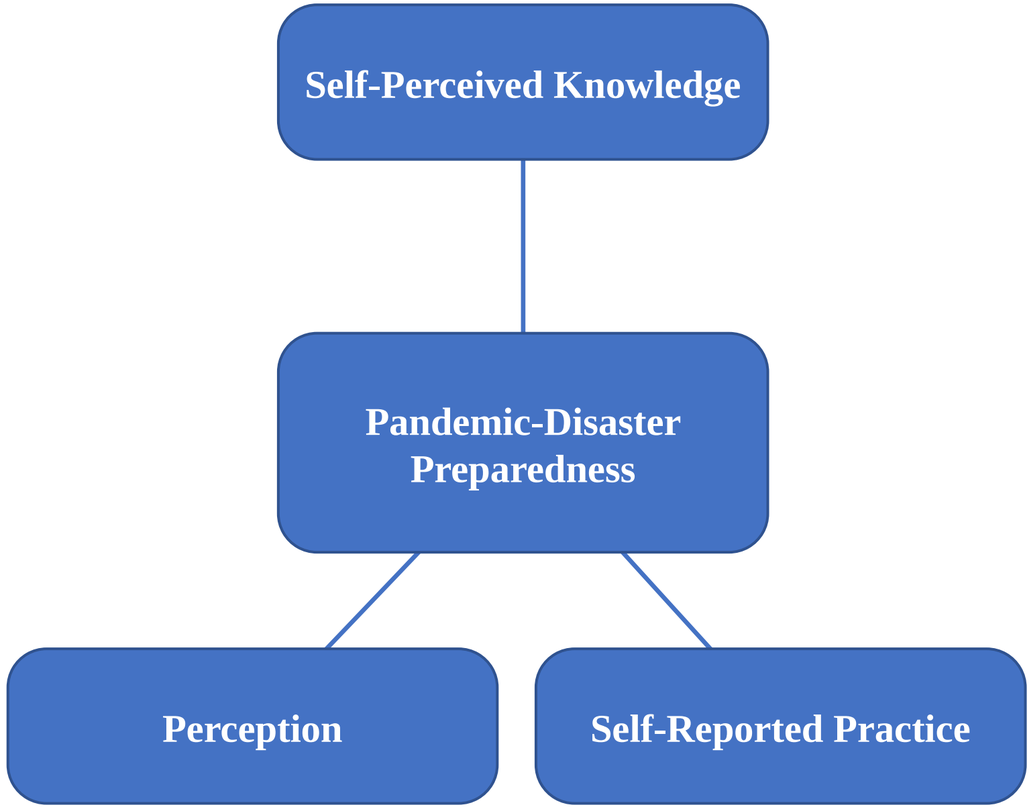


Figure 1. Conceptual framework on nurses' knowledge, perception and practice about pandemic disasters

Literature Review

The literature was reviewed around four themes, Saudi evidence and methodological limitations:

- Nurses' knowledge of pandemic and disaster preparedness
- Nurses' perceptions of risk, readiness and professional role
- Nurses' practices in pandemic disaster response
- Factors predicting knowledge, perception, practice and preparedness
- Saudi Arabian evidence and regional relevance
- Methodological and conceptual limitations of existing evidence

Literature Review

- A systematic review by Said and Chiang (2020) of 12 studies (1,443 nurses) found **inadequate disaster knowledge and skills** and **weak psychological preparedness**.
- Tas and Cakir (2022) reviewed 17 studies and found nurses **inadequately prepared**, while **disaster education and experience** were consistently linked to better results.
- Songwathana and Timalsina (2021) reported **poor-to-fair preparedness** in developing countries, with gaps in structured education, drills and disaster participation.
- Küçük et al. (2023) found only **moderate perceived preparedness** despite training and drill participation.

Literature Review

- Song et al. (2021) showed that **training, emergency drills and prior infectious-disease experience** were associated with stronger emergency-response competencies.
- Chegini et al. (2022) linked **training, advanced education and drill exposure** to higher preparedness; communication lagged behind technical competence.
- AlOtaibi et al. (2024) found **non-significant change** after a short online course, while Hung et al. (2021) reported **improvements** after an intensive mixed-format course.
- **Saudi studies** from Riyadh, Jeddah and Dammam used varying populations, concepts and instruments, giving no comprehensive regional picture.

Gap in Knowledge

Saudi evidence on nurses' pandemic preparedness is fragmented, and the only Taif study examined **emergency nurses' disaster-education needs** rather than pandemic-specific knowledge, perception and practice among staff nurses. Assessing a single domain cannot show where gaps lie. **No published study** has examined all three domains together among nurses in **governmental hospitals in the Taif Region**. This study provides **baseline data** for nursing education, simulation training and hospital emergency planning.

Methodology

Design

Quantitative, descriptive **cross-sectional** design.

01

Setting & Participants

Staff nurses in **governmental hospitals** in the Taif Region, Saudi Arabia.

02

Sample Size

G*Power 3.1 required **159**; **300 nurses** were included, plus 30 in the pilot.

03

Sampling

Purposive non-probability sampling technique.

04

Methodology

Inclusion criteria	Nurses who majored in nursing.
	Minimum six months of experience in hospitals in the Taif Region.
	Provided voluntary informed consent.
Exclusion Criteria	Student nurses and interns.
	Administrative staff.
	Nurses on long-term leave or who refused consent.

- **Sample distribution:** main study **300** + pilot study **30** (10%) = **330** nurses recruited; pilot participants were analyzed separately.

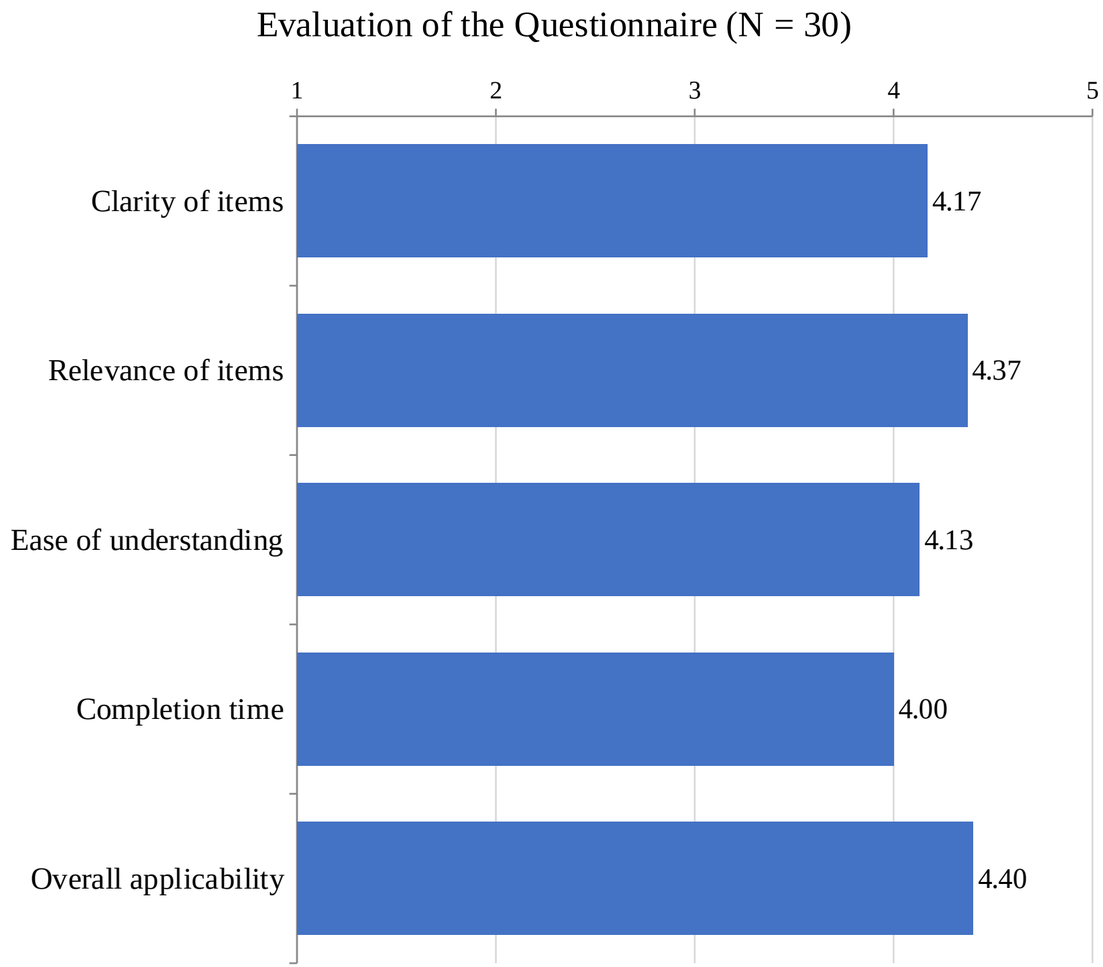
Measurement Instruments

- A structured self-report questionnaire was used. It had two parts:
- **Part I:** demographic sheet (age, gender, marital status, educational level, years of experience, practice area).
- **Part II:** 43-item questionnaire on nurses' disaster knowledge, attitudes and practices (Vishba & Shaheen, 2025): **15** knowledge, **14** perception and **14** practice items.
- 5-point Likert scale from 1 (**strongly disagree**) to 5 (**strongly agree**).
- Written permission obtained from the corresponding author (October 15, 2025).



Pilot Study

- **30 nurses** meeting the same criteria were excluded from the main sample.
- They rated clarity, relevance, ease of understanding, completion time and applicability (1 = poor, 5 = excellent).
- Mean ratings: **4.00 to 4.40**, supporting face validity and acceptability.
- Minor wording and formatting changes were made.



Data Collection Procedure

- Data were gathered after receiving **IRB approval** and the consent of the participating hospitals.
- Participants received **informed-consent forms** outlining the aims, procedures, risks and benefits.
- The questionnaire was completed **in person** during a scheduled shift or via a **secure online** link.
- Completed questionnaires were collected **confidentially** and kept locked in a secure place.

Ethical Considerations

- Approved by the **Institutional Review Board (IRB)**; informed consent obtained from all participants.
- **Beneficence:** research designed to benefit participants and disaster preparedness.
- **Justice:** fair and equitable treatment, free of exploitation and bias.
- **Respect for Human Dignity:** participants' free will and rights preserved.
- **Confidentiality:** data de-identified and stored in password-protected files.
- **Freedom to Withdraw:** at any time, without penalty.



Data Analysis

- Data were analyzed using **IBM SPSS Statistics, version 28**.
- **Descriptive:** frequencies, percentages, means, SD, ranges, skewness and kurtosis.
- **Reliability:** Cronbach's alpha for each domain.
- **Correlation:** Pearson correlations among the three domains.
- **Regression:** simple linear regression of self-perceived knowledge on perceptions and practice.
- **Group comparisons:** independent-samples t tests (gender); one-way ANOVA with Tukey post hoc tests (educational level); effect sizes reported (APA 7).

Sample Characteristics

Characteristic	Distribution
Female	61.7%
Male	38.3%
Age 20–30	24.0%
Age 31–40	23.0%
Age 41–50	25.7%
Age 51+	27.3%
Married	61.3%
Diploma	9.7%
BSN	85.0%
MSN	5.3%

Gender:

- Female nurses predominate (185; 61.7%).

Age:

- Evenly distributed; largest group 51+ years (27.3%).

Education:

- 85.0% hold a BSN; 9.7% diploma; 5.3% MSN.

Experience:

- Most had 1–5 years (32.7%); 23.3% had >15 years.

Practice area:

- Mainly pediatrics (34.0%) and ICU (23.3%).

Results – Psychometric Properties

Variable	Items	α	M	SD	Actual Range	Potential Range	Skewness	Kurtosis
Self-Perceived Knowledge	15	.91	61.64	4.64	47–73	15–75	−0.51	0.23
Perception	14	.88	57.69	4.45	43–68	14–70	−0.55	0.30
Self-Reported Practice	14	.73	57.25	5.37	31–68	14–70	−1.16	3.20

Reliability: Satisfactory internal consistency: α = **.73** (practice) to **.91** (knowledge).

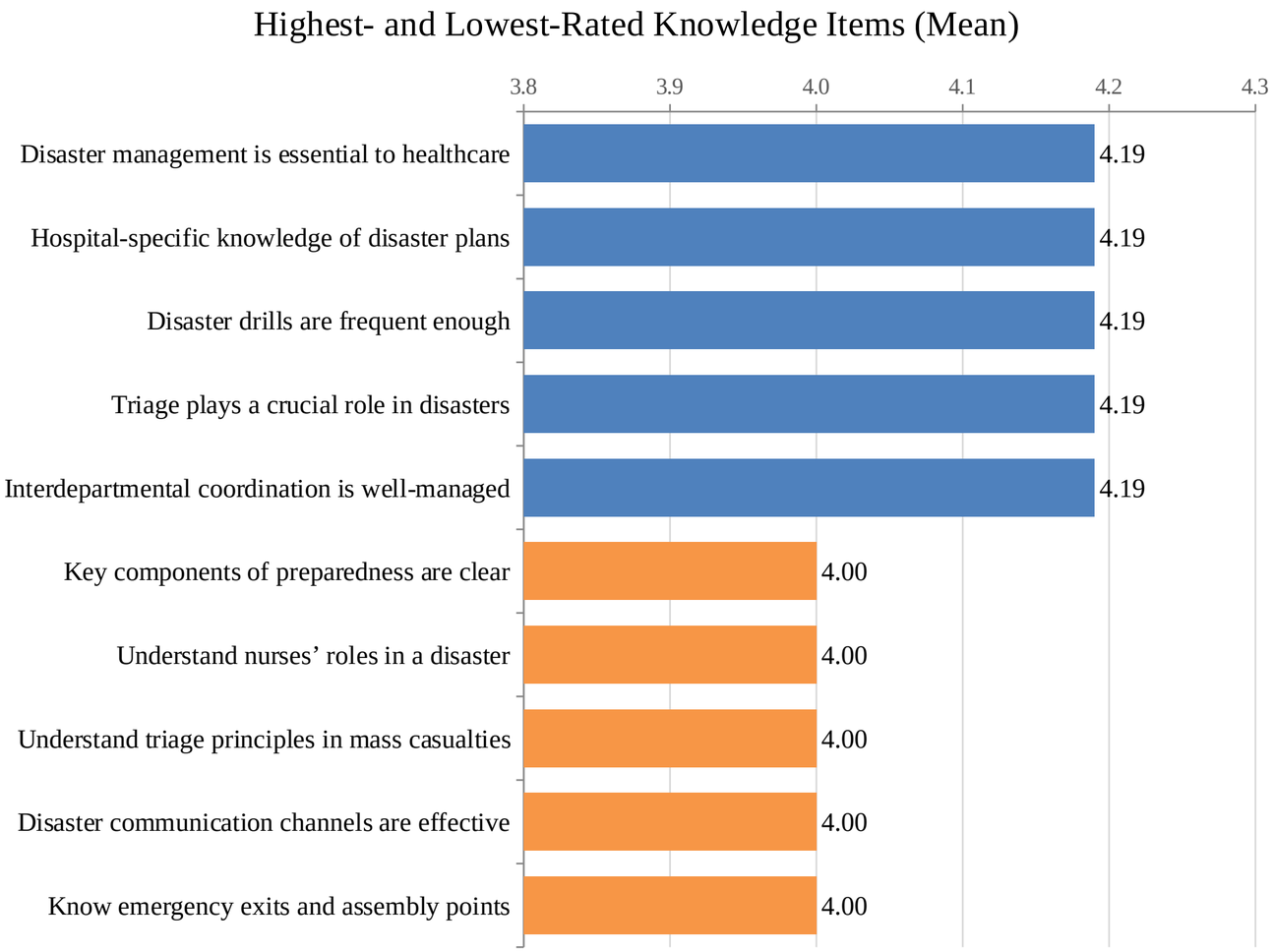
Distribution: Negative skewness in all scales: responses concentrated at the **higher end**.

Results – Self-Perceived Knowledge

Overall: **M = 61.64 (SD = 4.64)**, high level

Highest: Disaster management, hospital plans, drills, triage role and coordination (**4.19**).

Lowest: Key components, nurses' roles, triage principles, **communication channels**, emergency exits (**4.00**).

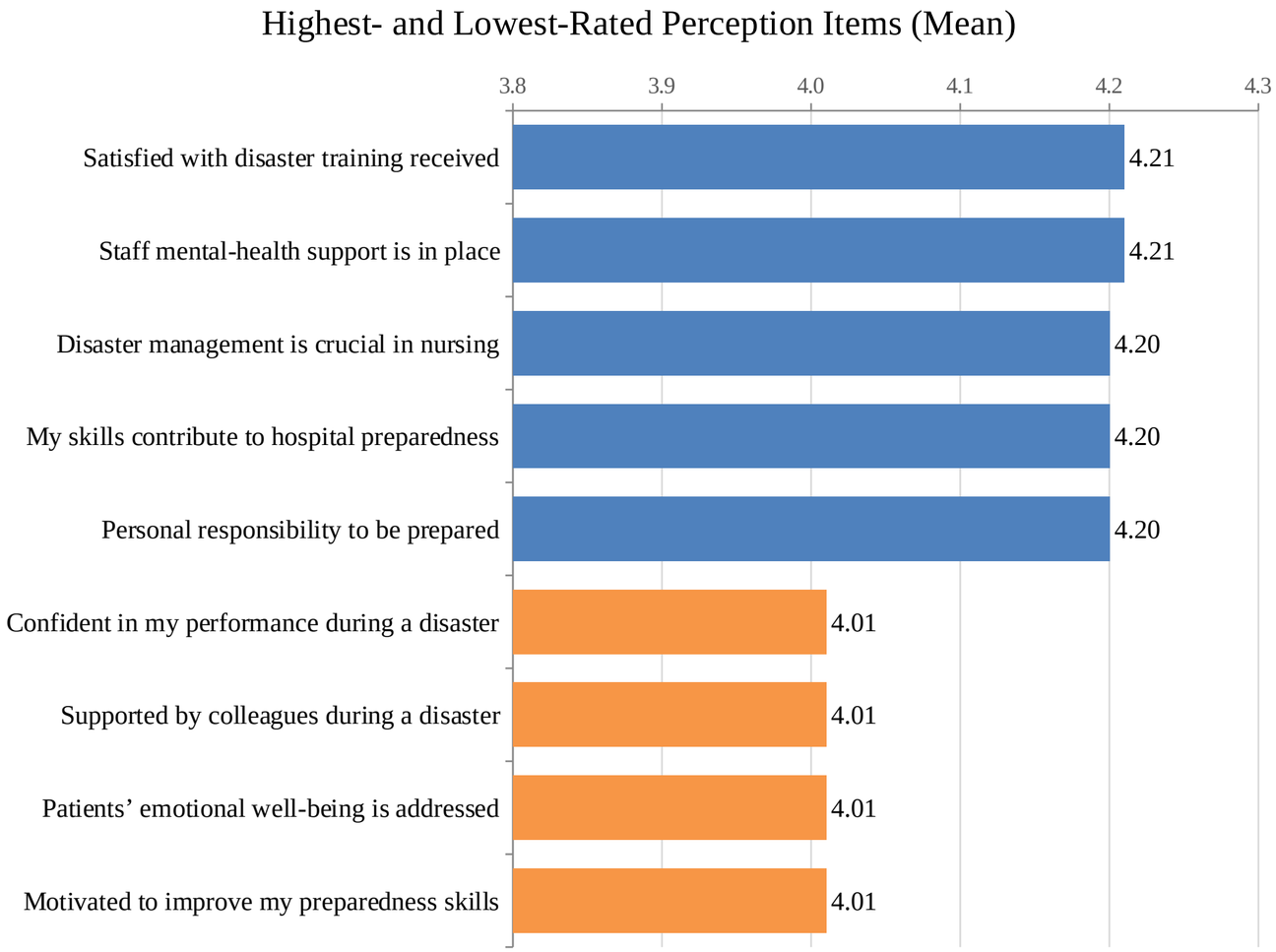


Results – Perceptions

Overall: **M = 57.69 (SD = 4.45)**, high level

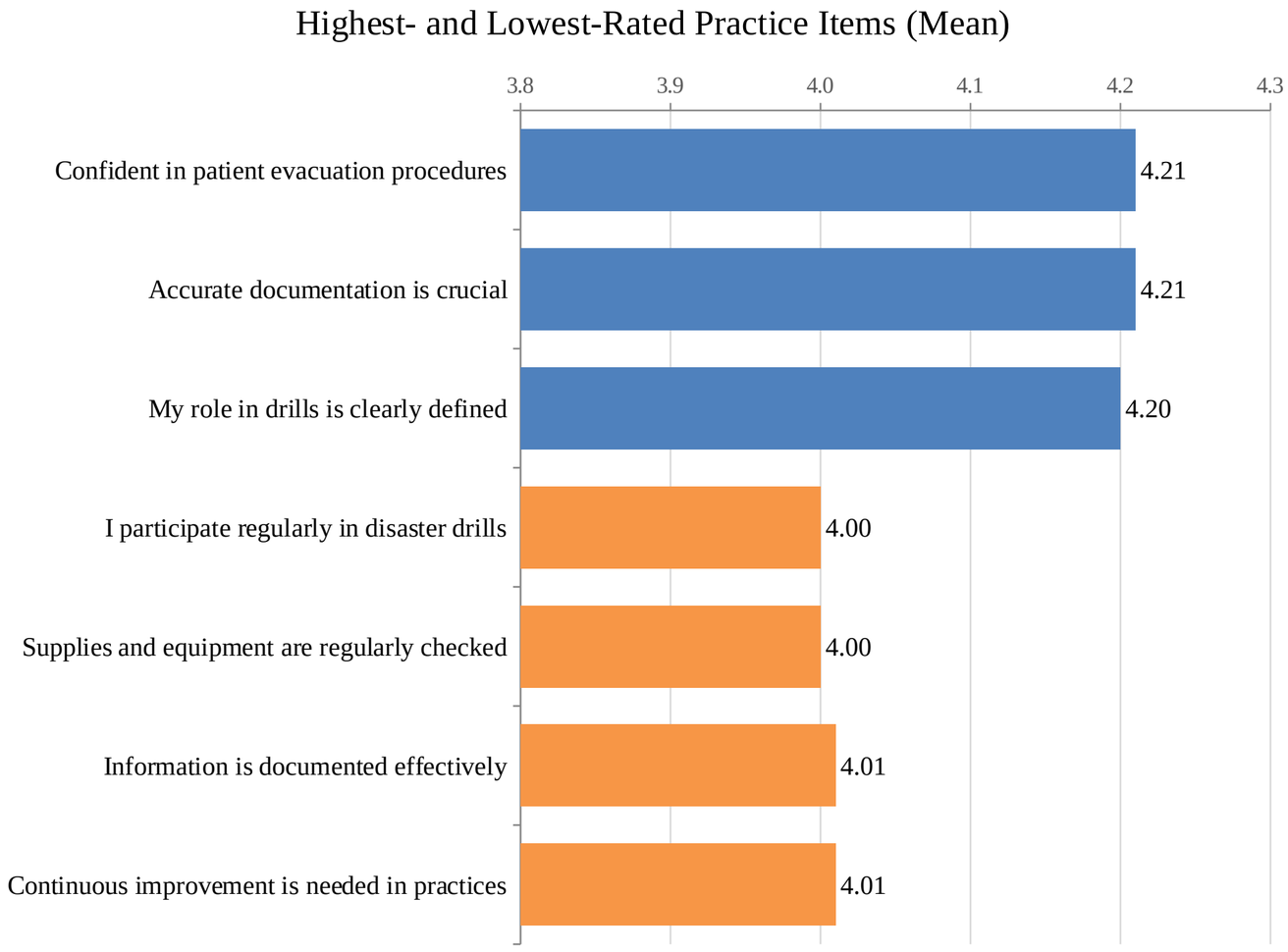
Highest: Satisfaction with training and staff mental-health support (**4.21**).

Lowest: Personal confidence, colleague support, **patients' emotional well-being** and **motivation** (**4.01**).



Results – Self-Reported Practice

Overall:	**M = 57.25 (SD = 5.37)**, high level
Highest:	Evacuation procedures and accurate documentation (**4.21**).
Lowest:	Regular drill participation and equipment checks (**4.00**); documentation and continuous improvement (**4.01**).



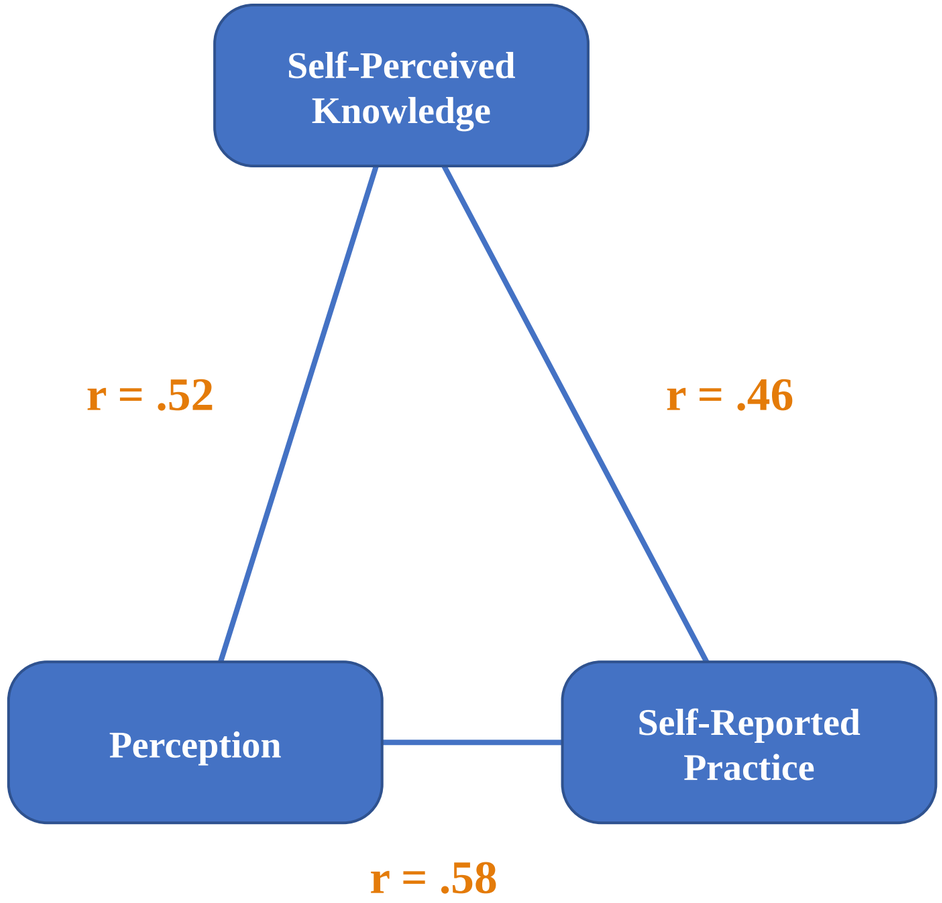
Results – Relationships

Pearson correlations among the three domains (N = 300)

Variable	M	SD	1	2	3
1. Self-Perceived Knowledge	61.64	4.64	—		
2. Perception	57.69	4.45	.52***	—	
3. Self-Reported Practice	57.25	5.37	.46***	.58***	—

*** $p < .001$

- All three domains were **positively and moderately** correlated.
- Strongest: **perception–practice** (r = .58).
- Correlations indicate **association, not causation**.

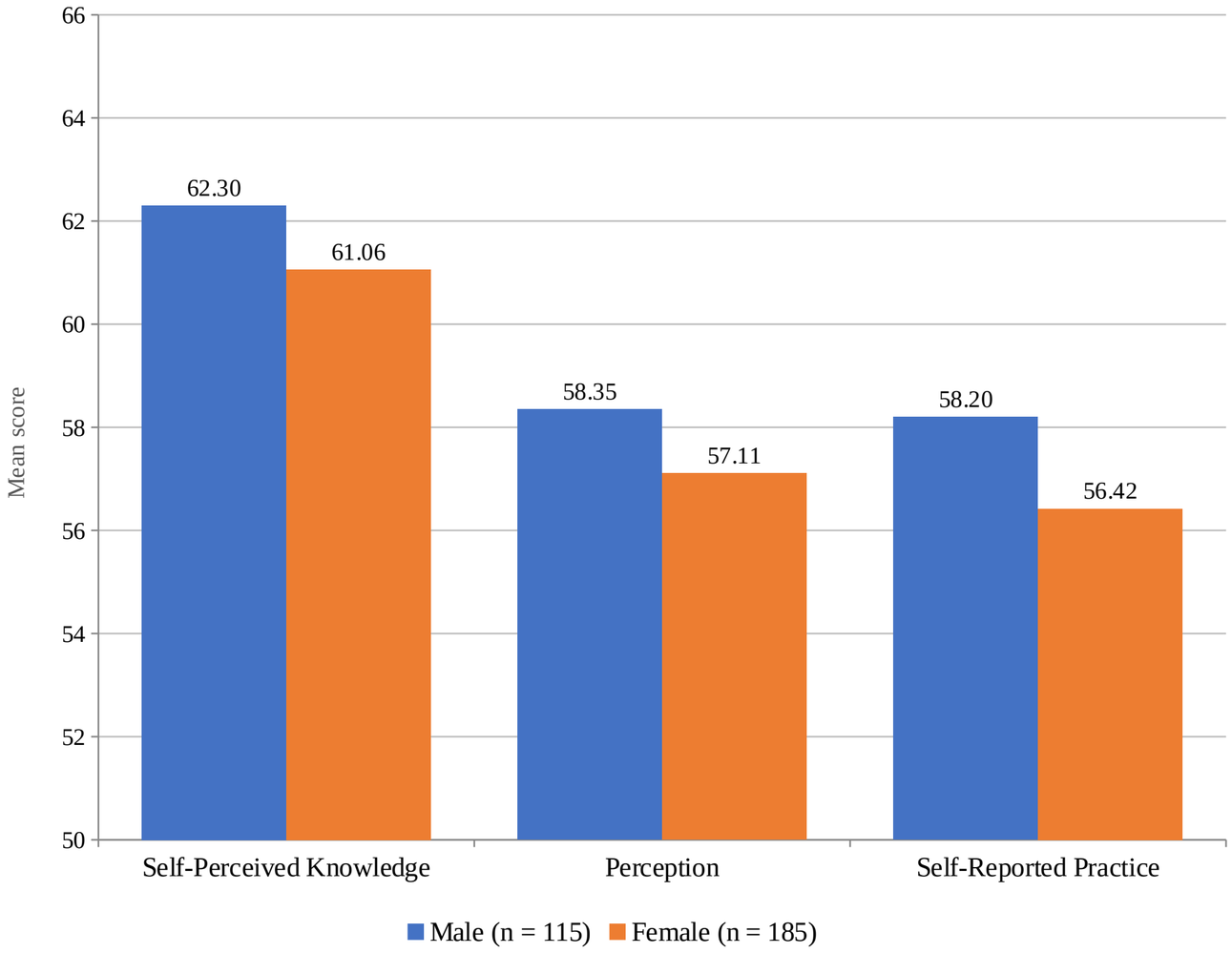


Results – Knowledge as a Predictor

Simple linear regression (N = 300)

Perception:	Knowledge explained **27.0%** of the variance ($R^2 = .270$, $\beta = .52$), $F(1, 298) = 110.45$, $p < .001$.
Practice:	Knowledge explained **21.2%** of the variance ($R^2 = .212$, $\beta = .46$), $F(1, 298) = 80.13$, $p < .001$.
Meaning:	Higher self-perceived knowledge predicted more positive perceptions and higher reported practice; this is **statistical prediction, not causation**.

Results – Differences by Gender



Knowledge:

- t(298) = 2.33, p = .021, d = 0.27

Perception:

- t(298) = 2.42, p = .016, d = 0.28

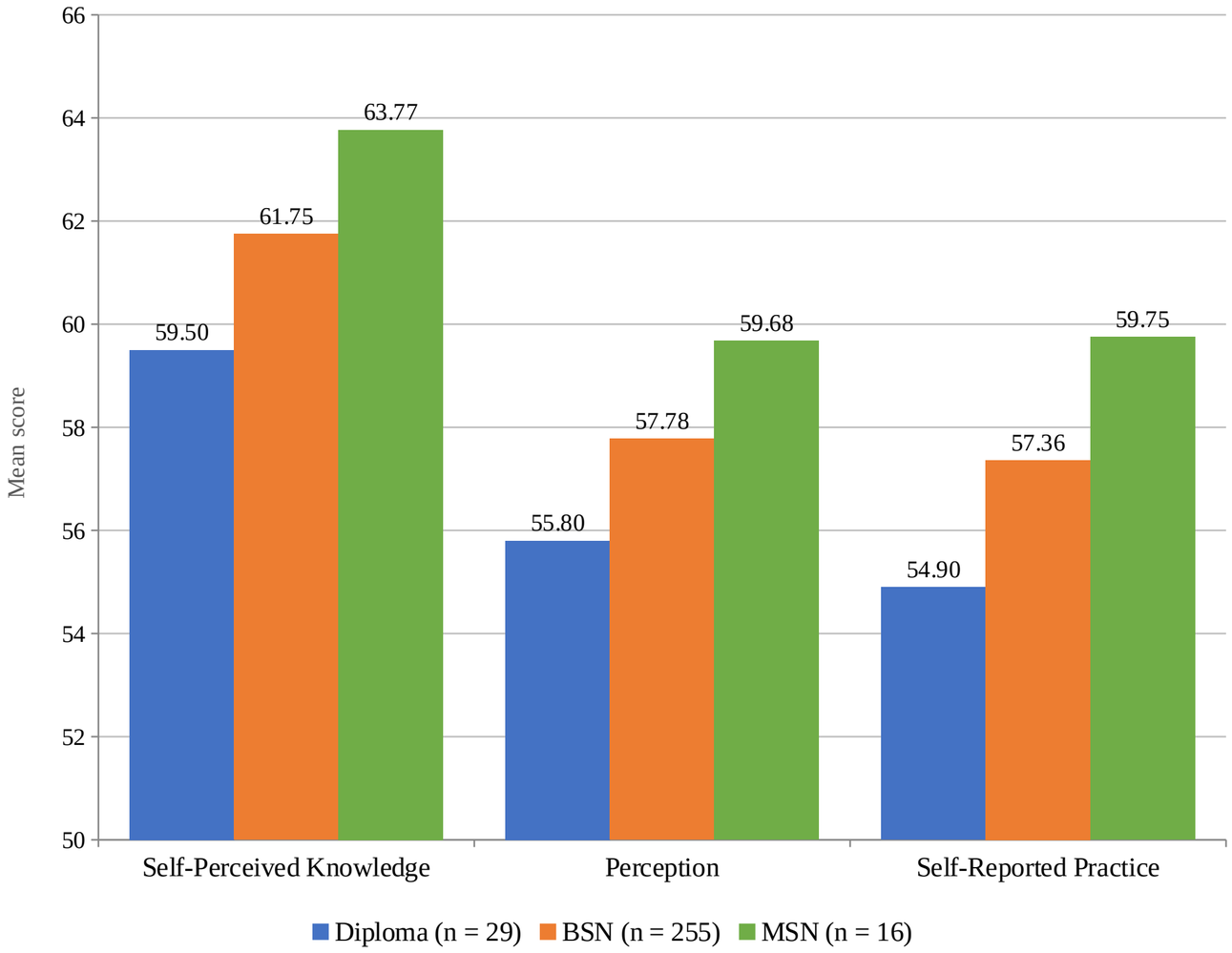
Practice:

- t(298) = 2.89, p = .004, d = 0.33

Overall:

- Male nurses scored higher in all three domains; **small effects**.

Results – Differences by Education



Knowledge:

- $F(2, 297) = 5.14$, $p = .006$, $\eta^2 = .033$

Perception:

- $F(2, 297) = 4.44$, $p = .013$, $\eta^2 = .029$

Practice:

- $F(2, 297) = 4.85$, $p = .009$, $\eta^2 = .032$

Tukey:

- MSN > Diploma in all three domains; **small effects**.

Discussion

- Nurses reported **high self-perceived knowledge** (M = 61.64).

Consistent with Younis et al. (2020), Saidam & Eljedi (2020) and Alrowili et al. (2025); reviews found insufficient or moderate knowledge (Said & Chiang, 2020; Tas & Cakir, 2022).

- Nurses held **positive perceptions** (M = 57.69).

In line with Saidam & Eljedi (2020) and Alan et al. (2022); Küçük et al. (2023) found only moderate perceived preparedness.

- Nurses reported a **high level of practice** (M = 57.25).

Aligned with Younis et al. (2020), Song et al. (2021) and Hong et al. (2022); Karnjuš et al. (2021) and Thirunavukkarasu et al. (2025) reported practice gaps.

Discussion

Relationships & Prediction

- **Moderate positive correlations** show the domains are related yet distinct; the strongest link was **perception–practice** (r = .58), similar to Thirunavukkarasu et al. (2025), Gandhi et al. (2021) and Alan et al. (2022).
- Knowledge explained **27.0%** of perception and **21.2%** of practice, consistent with Hong et al. (2022) and Song et al. (2021).
- **73.0% and 78.8%** of the variance remained unexplained, pointing to training, experience, organizational support, staffing, leadership, resources and resilience.

Discussion

Group Differences

Gender

- Small differences favoring male nurses may reflect **assignments, emergency exposure and training opportunities** rather than gender itself; evidence is mixed (Hong et al., 2022; Chegini et al., 2022).

Educational Level

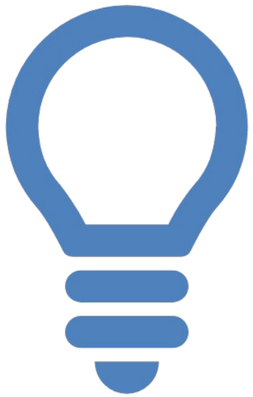
- **MSN-prepared** nurses scored higher than diploma-prepared nurses, consistent with Chegini et al. (2022), Song et al. (2021) and Hung et al. (2021); effects were small and the MSN group was small.
- Formal qualifications are **not the only route** to preparedness: experience, simulation and lifelong learning also build competency.

Discussion

Integrated Interpretation

- Pandemic-disaster preparedness is **multidimensional** and does not depend on knowledge alone.
- Nurses who felt more **confident, supported and responsible** reported more preparedness behaviors; the direction cannot be determined.
- Programs should combine theory with **drills, multidisciplinary exercises, communication readiness, psychological-skills training** and periodic competency testing.
- High scores are **self-assessed**, not proof of objective mastery; social desirability and COVID-19 exposure may have influenced responses.

Strengths and Limitations of the Study



- **Strengths of the Study**
  - Assessed knowledge, perception and practice together, with their relationships.
  - First assessment of its kind in Taif governmental hospitals; sample of 300 exceeded the minimum of 159.
  - Pilot-tested instrument with acceptable reliability (α = .73–.91).
- **Limitations of the Study**
  - Cross-sectional design: associations only, no causal conclusions.
  - Purposive sampling restricted to Taif governmental hospitals limits generalizability.
  - Self-reported data may involve recall, response and social-desirability bias.
  - Limited validity evidence; knowledge not objectively tested and practice not observed.
  - Unequal educational groups, especially the small MSN group.

Recommendations

1. Reinforce **simulation-based training** that mirrors real disaster conditions to improve decision making, communication and team coordination under pressure.
2. Strengthen and regularly practice common **communication protocols and chains of command** during training.
3. Provide **psychological support**: stress-management programs, counseling and resilience-building workshops.

Recommendations

4. Enhance **continuing professional development** with practical exercises, case-based learning and reflective practice.
5. Create a process of **periodic review and feedback** after disaster drills and actual incidents.
6. Policymakers should include **disaster preparedness in compulsory nursing competencies** at institutional and national levels.
7. Future research: **longitudinal, multicenter and experimental** studies; public and private hospitals; **probability sampling**; objective knowledge tests, observed practice and qualitative interviews.

Implications

The study presents several implications for **clinical practice, nursing education, hospital management and health policy**.

Nurses are at the core of pandemic response; their readiness shapes **patient outcomes, safety** and the performance of the emergency-response system.

Implications for Practice and Education

- **Practice**
  - Enhancing nurses' **communication, decision-making and psychological preparation** can improve readiness in clinical settings.
- **Education**
  - Integrate disaster management **holistically into nursing curricula**, with simulation exercises rather than theory alone.

Implications for Management and Policy

- **Management**
  - Provide **continuous preparedness training** with sufficient resources, time, institutional support and interdisciplinary collaboration.
- **Policy**
  - Develop **uniform national guidance** on disaster-preparedness training so that all nurses are prepared, whatever their setting.

Conclusions

Nurses in governmental hospitals in the Taif Region reported relatively **high self-perceived knowledge, positive perceptions and high self-reported practices** regarding pandemic disasters.

The three domains were **positively associated**, and self-perceived knowledge statistically predicted perceptions and self-reported practice. **Small but significant differences** were found by gender and educational level.

Emergency communication, psychological preparedness and ongoing review of protocols need further development through **continuous education, simulation and institutional support**.

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THANK YOU